

Report 11 Advanced Agents Simulation

Anomaly Satellite

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Report 11 — Advanced Agents: Simulation, Anomaly Detection & Satellite Discovery

Project: Jaljeev — Agentic Marine Intelligence System
Phase: Phase 0 (Advanced) — Items Originally Planned for Phase 2
Focus: Counterfactual simulation, SST anomaly detection, satellite data discovery

1. Simulation Agent — “What If?” (simulation_agent.py)

1.1 The Problem It Solves

A fisherman asks: “What if I leave at 8 AM instead of 5 AM?”
A sailor asks: “What if the wind picks up to 25 knots?”
A port operator asks: “What if I take a different route?”

These are **counterfactual questions** — they require re-running the entire risk pipeline with one or more parameters changed, then comparing the results.

1.2 Three Scenario Types

Type	User Changes	System Re-Runs
Time shift	Departure hour/day	Weather + ocean forecast at new time → full re-score
Condition override	Wave height, wind speed, etc.	Risk scoring with overridden variables
Vessel swap	Different vessel class	Same conditions, different thresholds and factors

1.3 Output: Delta Report

The Simulation Agent returns both the original and the modified assessment, plus a structured delta:

```
class SimulationResult(BaseModel):
    original: RiskAssessment
    modified: RiskAssessment
    delta_score: int           # +/- change
    delta_level: str           # e.g., "MODERATE → HIGH"
    changed_factors: list[str] # Which factors moved and by how much
    recommendation: str        # "Leaving at 8 AM increases risk by 15 points..."
```

1.4 Why This Is Innovation #4

“User changes departure/weather/vessel; system re-executes and returns delta — most memorable demo feature.”

Counterfactual reasoning is something no existing marine app offers. It demonstrates that the system doesn’t just report current conditions — it *reasons* about alternatives.

2. Anomaly Detection Agent (`anomaly_agent.py`)

2.1 The Problem It Solves

A researcher asks: “Why has fish productivity declined near Chilika in 2025?”

This requires comparing current conditions against a historical baseline and detecting statistically unusual values.

2.2 Two Detection Methods

Method	Library	Approach
Z-score	scipy	Deviation from climatological monthly mean. Simple, interpretable.
Isolation Forest	scikit-learn	Multivariate anomaly — detects unusual <i>combinations</i> of variables

2.3 SST Baseline: Honest About Its Limitations

The anomaly baseline is built from ERA5 SST reanalysis data, but currently only covers **a single year (2023)** — not a true multi-year climatology (which would require 30+ years for statistical robustness).

Standing rule: Every response from the Anomaly Agent explicitly labels its baseline as single-year:

“Note: This anomaly assessment uses a single-year (2023) SST baseline, not a multi-year climatology. Results should be interpreted as deviations from 2023 conditions, not from long-term normals.”

This caveat appears in: - The agent’s output schema (`baseline_note` field) - The evidence receipt - The LLM answer text (or template renderer) - The frontend EvidencePanel

2.4 Baseline Construction

Script: `scripts/build_sst_climatology.py` - Downloads ERA5 monthly SST for 2023 - Computes per-month mean and standard deviation for each grid cell - Stores as a processed Parquet file in `data/processed/` -
$$\text{Z-score} = (\text{current_sst} - \text{monthly_mean}) / \text{monthly_std}$$

3. Satellite Discovery Agent (`satellite_discovery_agent.py`)

3.1 The Problem It Solves

A researcher asks: “What satellite data is available for this region and time?”

The Satellite Discovery Agent searches NASA’s Common Metadata Repository (CMR) and STAC-compliant catalogs for relevant granules.

3.2 Capabilities

Feature	Implementation
Search by bbox + time range	CMR granule search via <code>pystac-client</code>
Filter by platform	EOS-06, Oceansat-2, SARAL/AltiKa, Sentinel-3
Filter by variable	SST, chlorophyll, ocean colour, altimetry
Return granule metadata	Coverage, acquisition time, cloud cover, download links

3.3 Integration with Other Agents

The Satellite Discovery Agent doesn't fetch or process the actual satellite data — that's the job of MOSDAC, Copernicus, and NASA OceanColor adapters. Instead, it answers the meta-question: "what observations exist?" This supports researchers and validates that the other adapters are pulling from available data.

4. Uncertainty Engine

4.1 Multi-Source Disagreement

When multiple sources report the same variable (e.g., SST from NOAA + Copernicus), the engine computes: - **Ensemble mean** — the consensus value - **Standard deviation** — the disagreement - **Flag** — raised if disagreement exceeds thresholds ($>1^{\circ}\text{C}$ for SST, $>0.3\text{ m/s}$ for currents)

4.2 Data Freshness Scoring

Each variable carries its source timestamp. The uncertainty engine flags: - **Stale** — data older than the maximum acceptable staleness (per PRD §7.2) - **Adequate** — within freshness bounds - **Real-time** — less than 1 hour old

4.3 Combined Confidence

```
confidence = "HIGH"    if disagreement < threshold and freshness == "adequate"
confidence = "MEDIUM"  if disagreement > threshold or freshness == "stale"
confidence = "LOW"      if multiple sources missing or data gaps
```

5. Phase 2 Items Landed in Phase 0

Item	Planned Phase	Delivered Phase
Simulation Agent (3 scenario types)	Phase 2	✓ Phase 0
Anomaly Detection Agent	Phase 2	✓ Phase 0
Uncertainty / Confidence Scoring	Phase 2	✓ Phase 0
Satellite Discovery Agent	Phase 1	✓ Phase 0
Validation / Critic Agent	Unplanned	✓ Phase 0

6. Complete Agent Roster (as of Aug 27, 2026)

#	Agent	Status
1	Weather Agent	✓ Done
2	Ocean Agent	✓ Done
3	Geo Agent	✓ Done
4	Risk Agent (two-stage)	✓ Done
5	Route Agent (radial scan)	✓ Done
6	Route Optimizer (A* on H3)	✓ Done
7	Passage Agent (port-to-port)	✓ Done
8	Simulation Agent	✓ Done
9	Anomaly Detection Agent	✓ Done
10	Satellite Discovery Agent	✓ Done
11	Validation / Critic Agent	✓ Done
12	Evidence Agent	✓ Done
13	Memory Agent	✓ Done
14	Fishing Agent	🔒 Blocked on MOSDAC PFZ archive

10 of 11 agents in the roster exist and run. The Fishing Agent is blocked on INCOIS PFZ archive access (the ML training labels).

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